



Syllabus-2024-2025

(SOET)(BTech-ComputerScience)

Title of the Course	Data structure and Algorithms in CSE/DS/CYBER
Course Code	CSL0309[T]

Part A

Year	2nd	Semester	3rd	Credits	L	T	P	C
					2	0	2	4
Course Type	Embedded theory and lab							
Course Category	Discipline Core							
Pre-Requisite/s	Basic understanding of computer fundamentals and programming in 'C'.			Co-Requisite/s				
Course Outcomes & Bloom's Level	CO1- Understanding: comprehensive knowledge of the data structures; (BL2-Understand) CO2- Applying: understand the importance of data and be able to identify the data requirements for an application; (BL3-Apply) CO3- Analyzing: have a practical experience of algorithmic design and implementation; (BL4-Analyze) CO4- Evaluating: practical experience of developing applications that utilize data structures and evaluating the performances of applications; (BL5-Evaluate) CO5- Creating: develop projects requiring the implementation of various data structures (BL6-Create)							
Course Elements	Skill Development ✓ Entrepreneurship ✗ Employability ✓ Professional Ethics ✗ Gender ✗ Human Values ✗ Environment ✗		SDG (Goals)	SDG1(No poverty) SDG2(Zero hunger) SDG4(Quality education) SDG8(Decent work and economic growth) SDG12(Responsible consupstion and production)				

Part B

Modules	Contents	Pedagogy	Hours
1	Introduction to Data Structures, Arrays: Linear Array, Operations on Linear Array, Multidimensional Array, Sparse Matrices; Strings; Linked List: Operations on Linked List, Garbage Collection, Header Linked List, Two-Way Linked List, Circular Linked List.	Real-world scenarios, Visual Aids, comparison chart, Interactive Example, and Experiments	12
2	Stacks: Implementation of Stacks using Arrays and Linked Lists, Polish Notations, Conversion from Infix to Postfix, Evaluation of Postfix Expressions; Queues: Representation of Queues, Implementation of Queues using Arrays and Linked Lists, Circular Queue, De-Queue, Priority Queues, Recursion, Tower of Hanoi problem.	Incremental Learning, Hands-On Practice, Real- World Applications, Visual and Interactive Tools, quiz, Group Discussions, and Experiments	8
3	Trees: Definition, Terminology; Binary Trees: Representation in Memory, Traversing Binary Tree, Extended Binary Tree, Threaded Binary Trees, Operations on Binary Trees; Search Trees: Operations on Search Trees.	Incremental Learning, Hands-On Practice, Real- World Applications, Visual and Interactive Tools, quiz, Group Discussions, and Experiments	10
4	Graphs: Terminology, Representation of Graphs, Directed Graphs, Directed Acyclic Graph, Shortest Path Algorithms, Graph Traversal; Minimum Cost Spanning Tree: Kruskal's Algorithm, Prim's Algorithm.	Real-world scenarios, Incremental Learning, Hands-On Practice, Real- World Applications, Visual and Interactive Tools, quiz, Group Discussions, and, Experiments	5
5	Internal and External sorting: Bubble Sort, Heap Sort, Radix sort, Selection sort, Insertion sort and Quick Sort; Searching: Linear Search, Binary Search. File Structures, Random Access Files, Indexed Sequential Files, B Trees and B+ Trees; Hash Tables, Hashing Functions, Collision Resolution Strategies.	Real-world scenarios, Incremental Learning, Hands-On Practice, Real- World Applications, Visual and Interactive Tools, quiz, Group Discussions, and,, Experiments	5

Part C

Modules	Title	Indicative-ABCA/PBL/ Experiments/Field work/ Internships	Bloom's Level	Hours
1-2	Identify problem, choose data structure, explain solution steps, analyze time complexity, test with examples, and discuss results.	Case Study	BL3-Apply	3
3-4	Define project goals, select relevant data structures, implement algorithms, test with real data, optimize, and document learning outcomes.	PBL	BL6-Create	2
1-5	In experiments, implement algorithms, test with various inputs, measure efficiency, analyze results, compare performance, and document observations systematically.	Experiments	BL3-Apply	4

Part D(Marks Distribution)

Theory					
Total Marks	Minimum Passing Marks	External Evaluation	Min. External Evaluation	Internal Evaluation	Min. Internal Evaluation
100	40	60	18	40	
Practical					
Total Marks	Minimum Passing Marks	External Evaluation	Min. External Evaluation	Internal Evaluation	Min. Internal Evaluation
100	50	60	30	40	

Part E

Books	Lipschutz, S. (2018). Schaum's Outline of Linear Algebra. McGraw-Hill. Kruse, R. L., Ryba, A. J., & Leung, B. P. (2006). Data Structures and Program Design in C.
Articles	
References Books	Horowitz, E., Sahni, S., & Anderson-Freed, S. (1981). Fundamentals of Data Structures in C. Publisher. reTmbley, J. P., & Sorenson, P. G. (2014). An Introduction to Data Structures with Applications.
MOOC Courses	
Videos	

Part F

Project 1	view
Project 2	view
Project 3	view
Project 4	view
Project 5	view

Course Articulation Matrix

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	-	-	-	-	-	-	-	-	-	-	3	2	2
CO2	2	3	1	-	-	-	-	-	-	-	-	2	2	2	1
CO3	2	3	1	-	-	-	-	-	-	-	-	2	2	3	1
CO4	3	3	1	-	-	-	-	-	-	-	-	2	2	3	1
CO5	2	1	-	-	-	-	-	-	2	-	-	2	3	2	2
CO6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-